

M.E EDUCATION CENTRE

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CHEMISTRY: 12 - HALF-YEARLY EXAM: 2025-26

SECTION "A" (SHORT QUESTIONS / ANSWERS)

NOTE: Attempt any FIVE questions from Inorganic and FIVE questions from Organic Chemistry. (10x3=30)

INORGANIC CHEMISTRY

- (i) Give reasons for the following:
(a) Why do transition elements show variable oxidation state?
(b) Why Cu^{+2} ion is blue but Zn^{+2} ion is colourless?
- (ii) Write down the balanced chemical equation for the following reactions:
(a) Reaction of permanganate with oxalic acid
(b) Reaction of Iron with Chlorine
- (iii) Write down the names of steel types and applications?
- (iv) Write the IUPAC name of the following:
(a) $[\text{Cr}(\text{en})_3](\text{NO}_3)_3$ (b) $[\text{Pt}(\text{NH}_3)_2\text{Cl}]_2^{2+}$ (c) $\text{Na}_3[\text{Co}(\text{C}_2\text{O}_4)_3]$
- (v) What do you mean by Chelates and Chelating agent? Give an Example
- (vi) What is flame test? Mention the color of flame of alkali metals.
- (vii) Write down the unique behavior of Beryllium in group II-A.

Steel
tool
stainers

ORGANIC CHEMISTRY

- (i) Write the uses and properties of Lignite and Anthracite?
- (ii) Explain the partial and total synthesis?
- (iii) Explain the reforming of petroleum?
- (iv) Give methods of preparation of Alkenes?
- (v) Give hydration, epoxidation and ozonolysis in ethene?
- (vi) Give acidity of terminal alkynes with two examples?
- (vii) What do you mean by Chiral centre? Give an example

SECTION "B" (LONG QUESTIONS / ANSWERS)

NOTE: Attempt any TWO questions from Inorganic and TWO questions from Organic Chemistry. (4x5=20)

INORGANIC CHEMISTRY

- (i) Describe the manufacturing of H_2SO_4 by the Contact Process.
- (ii) Describe the manufacture of Caustic Soda by the Castner-Kellner Cell
- (iii) Describe the various steps involved in the extraction of 99.99% pure copper from its chalcoppyrite ore.

ORGANIC CHEMISTRY

- (i) Explain the molecular orbital treatment of benzene
- (ii) What are organometallic compounds? How is Grignard reagent prepared? Write the reactions of CH_3MgI with Water, Alcohol and Ester.
- (iii) Write chemical equations of the following:
* Reaction of ethanol with acetic acid in the presence of H_2SO_4 .
* Reaction of phenol with concentrated HNO_3 .
* Reaction of alcohol with concentrated H_2SO_4 .
* Reaction of ethyle bromide with alcoholic KOH
* Reaction of benzene with fuming sulphuric acid
- (iv) What is meant by a nucleophile? Give the mechanism of the following reactions:
(a) SN^1 reaction between a 3° alkyl halide and NaCN .
(b) SN^2 reaction between a 1° alkyl halide and NaOH .